

SUBHANKAR DEY

<https://sites.google.com/view/subhankardeymath>

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RESEARCH INTERESTS

Floer homological invariants and their impact on several questions in knot theory, 3-manifold topology and contact topology.

ACADEMIC POSITIONS

Indian Institute of Technology Palakkad

Assistant Professor Gr. 1

Kerala, India

January, 2025 -

Ruhr-Universität Bochum

Postdoctoral Research Assistant

Mentor: **Dr. Claudius Zibrowius**

North Rhine-Westphalia, Germany

August, 2024- December, 2024

Durham University

Postdoctoral Research Assistant

Mentor: **Dr. Claudius Zibrowius**

Durham, UK

August, 2023- July, 2024

University of Alabama

Postdoctoral Assistant Professor

Mentor: **Dr. Bülent Tosun**

Alabama, USA

August, 2021- July, 2023

EDUCATION

University at Buffalo, State University of New York

PhD, Mathematics

Advisor: **Dr. Çağatay Kutluhan**

NY, USA

2015-2021

Ramakrishna Mission Vivekananda University (RKMVERI)

Master of Science, Mathematics

West Bengal, India

2013-2015

Ramakrishna Mission Vidyamandira

Bachelor of Science, Mathematics Honors

West Bengal, India

2010-2013

FELLOWSHIPS/AWARDS

- AMS-Simons Travel Grant, 2022 - 24 (\$2500 each year for research related travels, for two years)
- Mathematics Department Doctoral Dissertation Fellowship, University at Buffalo, SUNY, Spring 2020
- GSA Travel Fund, University at Buffalo, Fall 2019
- Junior Research Fellowship via National Eligibility Test, Council of Scientific & Industrial Research, Government of India, 2014

- NBHM Fellowship, National Board of Higher Mathematics, India, 2013 - 15
- Among top 25 candidates in MADHAVA Mathematics Competition (A Mathematics competition for undergraduate students across India), 2012
- Innovation in Science Pursuit for Inspire Research (INSPIRE), Govt. of India, 2010 - 13

PUBLICATIONS

Submitted/Published/To Appear:

- *Cable knots are not thin*: arXiv:1904.11591 (Submitted)
- *Combinatorial Description of epsilon invariant for knots in S^3* (Joint with **Hakan Doğa**): (in **Journal of Knot Theory and its Ramifications**, Vol. 30, No. 6 (2021))
- *Rank Bounds in Link Floer homology and Detection Results* (Joint with **Fraser Binns**): (In **Quantum Topology**, DOI 10.4171/QT/197)
- *Cable Links, Annuli and Sutured Floer homology* (Joint with **Fraser Binns**): arXiv:2207.08035 (Accepted in **Mathematical Research Letters**)
- *Unknotted curves on genus one Seifert surfaces of Whitehead doubles* (joint with **Veronica King***, **Colby Shaw***, **Bülent Tosun** and **Bruce Trace**): (In **Pacific Journal of Mathematics**, Vol. 330 (2024), No. 1, 123156)

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- *Floer homology, clasp braids and detection results* arXiv:2405.11224 (joint with **Fraser Binns**) (Submitted)
- *Algorithms in 4-manifolds topology* arXiv: 2411.08775 (joint with **Et. al**) (Collaborative work from the conference: '4-Manifolds and Algorithms' in University of Regensburg, September 2024) (Submitted)

In Preparation:

- *On Fractional Dehn Twist Coefficient of hyperbolic fibered slice knots* (joint with **Aprataim Chakraborty**)
- *Sutured Floer homology and detection of tangles* (joint with **Fraser Binns**, **Claudius Zibrowius**)
- *A note on homology concordance and generalized Mazur patterns*
- *Minors of Alexander matrix of links, and decategorification of sutured Floer homology* (joint with **Fraser Binns**, **Miriam Kuzbary**):
- *Khovanov homology, Link Floer homology and geography questions about Legendrian links* (joint with **Fraser Binns**)

TALKS

Department Colloquium, Harishchandra Research Institute, April 2025

Knots, quivers and beyond, IIT Bombay, February 2025

^{1*} are undergraduate students, part of *REU 2022: Topology at UA*

Department Symposium, IIT Palakkad, January 2025
 4-manifolds and Algorithms, University of Regensburg, September 2024
 Low dimensional topology days, University of Nantes, August 2024
 Short Talks Session, Institute Fourier Summer School, University of Grenoble-Alpes, June 2024
 Research Talk (Virtual), Department of Mathematics, IIT Roorkee, May 2024
 Geometry and Topology Seminar, Durham University, March 2024
 Research Talk (Virtual), Department of Mathematics, IIT Palakkad, January 2024
 TCG Crest IAI Seminar, Kolkata, June, 2023
 University of Oregon Topology/Geometry Seminar, May, 2023
 Special Session on *Contact and Symplectic Topology in dimensions 3 and 4*, AMS Spring Southeastern Sectional Meeting, Atlanta, March 2023
 Special Session on *Interactions between 3-manifolds and 4-manifolds*, AMS Fall Southeastern Sectional Meeting, Tennessee, October 2022
 UB Geometry and Topology Seminar, University at Buffalo, March 2022
 Georgia Tech Geometry and Topology Seminar, November 2021
 Special Session on *Low dimensional Symplectic and Contact Topology, Their Invariants and Interactions*, AMS Fall Southeastern Sectional Meeting, Alabama, November 2021
 Algebra-Topology Seminars, Department of Mathematics, University of Alabama, November 2021
 Binghamton University Graduate Conference in Algebra and Topology (BUGCAT-2020) (Virtual) November 2020
 Contributed Paper Session, AMS Fall Western Sectional Meeting (Virtual), October 2020
 Graduate Students Symplectic Reading Group Seminar, Department of Mathematics, Duke University, March 2020
 Graduate Students G&T Seminars, Department of Mathematics, Duke University, March 2020
 Geometry and Topology Seminar, Department of Mathematics, Duke University, March 2020
 DMS Seminar, Indian Institute of Science Education and Research (IISER) Kolkata, January 2020
 Binghamton University Graduate Conference in Algebra and Topology (BUGCAT-2019), Binghamton University, November 2019
 Special Session on *Invariants of Knots, Links and Low-dimensional Manifolds*, AMS Eastern Sectional Meeting, Binghamton, October 2019
 Geometry and Topology Seminar, University at Buffalo, October 2019
 Graduate Students Lecture Series, UB SUNY, Spring, Fall 2019

TEACHING AND MENTORSHIP EXPERIENCES

IIT Palakkad:

Instructor:

Probability and Statistics (MA 2041) (Spring 2025)

Linear Algebra (MA 5001) (Fall 2025)

University of Alabama:

Instructor:

Introduction to Linear Algebra (Math 237) (Spring 2023)

Discrete Mathematics (Math 301) (Spring 2022, Fall 2022)

Theory of Probability (Math 355) (Fall 2021)

Mentor:

Mentor of *REU 2022: Topology at UA* on knot theory and 3- and 4- dimensional manifolds, along with Dr. Bülent Tosun, University of Alabama, Summer, 2022

University at Buffalo:**Instructor:**

College Calculus 1 (MTH 141) (Online) (Summer 2021)

College Calculus 3 (MTH 241L) (Online) (Spring 2021)

Survey of Calculus and its Applications 2 (MTH 122L) (Online) (Fall 2020)

Survey of Calculus and its Applications 2 (MTH 122ML) (Online) (Summer 2020)

Teaching Assistant:

Survey of Multivariable Calculus (MTH 417) (Fall 2019)

College Calculus 2 (MTH 142) (Summer 2019)

Elements of Geometry (MTH 335) (Spring 2019)

Survey of Calculus and its Applications 2 (MTH 122) (Fall 2018, Fall 2020)

Survey of Calculus and its Applications 1 (MTH 121) (Summer 2018)

Introduction to Topology (MTH 427) (Fall 2017)

Introduction to Contemporary Mathematics (MTH 101) (Fall, 2017)

College Calculus 1 (MTH 141) (Summer 2017, Spring 2018)

Introduction to Linear Algebra (MTH 309) (Spring 2017)

College Calculus 3 (MTH 241) (Fall 2016, Fall 2018, Spring 2019, Spring 2021)

Mentor:

Led a semester long *Directed Reading Program*, with Connor Wilson, a Mathematics major undergraduate students, on *A hands-on comparative study of Euclidean and Hyperbolic Geometry*, (Fall, 2019)

Led a semester long *Directed Reading Program*, with two Mathematics major undergraduate students: Cameron Watt and Kwan Bo Shim, on *How to distinguish a ball from a doughnut...mathematically* (Spring, 2019)

Led a short *Directed Reading Program*, with Karrar Aljanahi, a Mathematics major undergraduate student, on *How to distinguish a ball from a doughnut...mathematically* (Winter, 2019)

CONFERENCES AND WORKSHOPS ATTENDED

Knots, quivers and beyond, IIT Bombay, February 2025

4-manifolds and algorithms, University of Regensburg, September 2024

Low dimensional topology days, University of Nantes, August, 2024

Institut Fourier Summer School: Low dimensional topology, University of Grenoble-Alpes, June 2024

Myhill Lecture Series (Dr. Tomasz Mrowka), University at Buffalo, April 2024

Interactions between Symplectic and Holomorphic Convexity in 4 Dimensions (virtually), BIRS Banff, April 2023

AMS Spring Southeastern Sectional Meeting, Atlanta, March 2023

Tech Topology Conference, GeorgiaTech, December 2022

Floer homotopical methods in low dimensional and symplectic topology (virtually), MSRI, November 2022

3rd Upstate New York Topology Seminar (UNYTS 2022), Syracuse University, October 2022

AMS Fall Southeastern Sectional Meeting, Tennessee, October 2022

Connections Workshop: Floer Homotopy Theory (virtually), MSRI, September 2022

Braids in Low Dimensional Topology, ICERM, April 2022

Tech Topology Conference (Virtual), December 2021

AMS Fall Southeastern Sectional Meeting (Virtual), November 2021

Joint Mathematical Meetings (Virtual), January, 2021

Workshop of Link Homology and Link Concordance (Virtual), November 2020

Binghamton University Graduate Conference in Algebra and Topology-2020 (Virtual), Binghamton University, November 2020

AMS Fall Western Sectional Meeting (Virtual), October 2020

Nearly Carbon Neutral Geometric Topology Conference (Virtual), June 2020, 2021

Trends in Low-Dimensional Topology (Virtual), April - July & Fall 2020

Visiting Student, Duke University, February 2020

Binghamton University Graduate Conference in Algebra and Topology-2019, Binghamton University, November 2019

AMS Eastern Sectional Meeting, Binghamton, October 2019

Perspectives on Dehn Surgery, ICERM, July 2019

Midwest Topology Seminar, Michigan, 2019

Virginia Topology Conference, Virginia, December 2018

The topology and geometry of low-dimensional manifolds: a celebration of the mathematics of Bob Gompf, Texas, Summer 2018

Low dimensional Topology Conference, Los Angeles, California, 2018

Topology Day, University at Buffalo, SUNY, New York, USA, 2017

Graduate Students Topology and Geometry Conference (GSTGC) in Indiana University, Indiana, USA, 2016

Workshop on Differential Topology in North-Eastern Hill University, Shillong, India, 2015

Seminar series on Topology and Geometry in Ramakrishna Mission Vivekananda University, Howrah, India, 2014

OUTREACH, SERVICE AND PROFESSIONAL ACTIVITY

Dissertation Committee Internal Member: Tamoghna Kar (2025-), IIT Palakkad

Reviewer: *zbMATH Open* ($\times 5$)

Referee: *Algebraic and Geometric Topology, Michigan Mathematical Journal, Quantum Topology, K3 problem list*

Co-organizer of *Yuktibhasa Seminar* series at the Department of Mathematics, IIT Palakkad, Spring 2025

Co-organizer of special session on *Heegaard Floer homology, related invariants and their applications* in **Nearly Carbon Neutral Geometric Topology Conference–2023**, along with Fraser Binns, June 2023

Co-organizer of *REU 2022: Topology at UA* on knot theory and 3- and 4- dimensional manifolds, along with Dr. Bülent Tosun, University of Alabama, Summer, 2022

Co-organizer of weekly *Algebra-Topology Seminars* at the Department of Mathematics, University of Alabama with Dr. Bülent Tosun. (Fall 2021-22, Spring 2022-23)

Organizer of *Graduate Student Lecture Series*, Department of Mathematics, UB SUNY (Fall, 2019 - 2020; Spring, 2020 - 2021)

Tutor in mathematics help center at Department of Mathematics, University at Buffalo (August 2015 - August 2020)

PROFESSIONAL DEVELOPMENT

Summer Trisectors Workshop (Virtual), Organised in partial by University of Nebraska-Lincoln, June 2021

Topology Students Workshop (Virtual), Organised by Georgia Institute of Technology, July 2020

Associate-level Certification: Centre for the Integration of Research, Teaching and Learning (CIRTL, USA), August 2020

Bring an Inclusive Mindset to Your Teaching, CIRTL (USA) Workshop, September 2020

SOCIAL OUTREACH

Former **Secretary** (2017 - 2020), current **Chair** (2022 - 2024), a Board of Directors member (2020 - 2024) and a mentor (2016 - present) in **Padakshep** (www.padakshep.org), a non-profit organization (licensed in the USA and India), which each year, takes care of education of a number of (15-20 students each year) financially needy but meritorious students through mentoring and financial assistance, belonging to remote areas in West Bengal, India.

Volunteer for Department of Mathematical Sciences in the Celebrate Science program at Durham, UK (2023).

LANGUAGE

Bengali (Native), Hindi (Fluent), English (Fluent)